PWM Generation for Motor/ESC Control Using STM32

This project demonstrates how to generate a PWM signal using the STM32 Timer peripheral to control a motor or ESC.

The PWM signal is generated using the STM32 HAL library and configured through STM32CubeMX/ STM32CubeIDE.

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1. STM32CubeMX / IOC Configuration

Select Timer

Open the `.ioc` file in STM32CubeMX or STM32CubeIDE.

Select the required Timer peripheral.

Example:

```
TIM2
```

Configure Clock Source

Select the appropriate clock source according to the STM32 board and clock configuration.

```
Clock Source → Internal Clock / External Clock
```

Configure PWM Channel

Select the required timer channel and configure it for PWM generation.

Example:

```
Channel → PWM Generation CH1
```

For example:

```
TIM2 → Channel 1 → PWM Generation CH1
```

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2. Timer Parameter Settings

For a typical ESC control signal, a PWM frequency of **50 Hz** is commonly used.

The PWM period is calculated using:

```
T = 1 / f
```

For 50 Hz:

```
T = 1 / 50
T = 0.02 seconds
T = 20 ms
T = 20,000 µs
```

If the timer counter is configured with a **1 µs timer tick**, the Counter Period is:

```
Counter Period = 20,000 - 1
               = 19,999
```

Therefore, the timer configuration can be:

```
Prescaler       → Timer Clock / 1 MHz - 1
Counter Mode    → Up
Counter Period  → 19999
Auto Reload     → Enable
PWM Mode        → PWM Mode 1
Pulse           → Initial PWM pulse width
```

For example, a typical ESC may use:

```
1000 µs → Minimum throttle
1500 µs → Mid throttle
2000 µs → Maximum throttle
```

> **Note:** The exact PWM pulse range depends on the ESC. Some ESCs may require calibration before operation.

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3. PWM Generation in `main.c`

After configuring the timer in STM32CubeMX, start PWM generation using:

```
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);
```

This starts PWM output on:

```
TIM2 Channel 1
```

The timer and channel depend on the configuration selected in the `.ioc` file.

Changing PWM Pulse Width

The PWM pulse width can be changed using:

```
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, VALUE);
```

For example:

```
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 1000);
```

If the timer tick is configured to **1 µs**, this produces approximately:

```
1000 µs pulse width
```

Similarly:

```
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 1500);
```

produces approximately:

```
1500 µs pulse width
```

---

4. Motor/ESC Speed Variation

The PWM pulse width can be gradually increased to change the motor/ESC throttle command.

Example:

```
for (int i = 1000; i <= 2000; i += 100)
{
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, i);

    HAL_Delay(3000);
}
```

The PWM pulse width changes as follows:

```
1000 µs
    ↓
1100 µs
    ↓
1200 µs
    ↓
1300 µs
    ↓
1400 µs
    ↓
1500 µs
    ↓
1600 µs
    ↓
1700 µs
    ↓
1800 µs
    ↓
1900 µs
    ↓
2000 µs
```

Each PWM value is maintained for approximately **3 seconds**.

To return the motor/ESC command to minimum throttle:

```
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 1000);
```

Complete Example

```c
/* Start PWM generation */

HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);

while (1)
{
    for (int i = 1000; i <= 2000; i += 100)
    {
        /* Change PWM pulse width */

        __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, i);

        /* Wait 3 seconds */

        HAL_Delay(3000);
    }

    /* Return to minimum throttle */

    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 1000);

    HAL_Delay(3000);
}
```

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5. Verify PWM Configuration

If PWM output is not generated, check the following.

Open:

```
stm32f4xx_hal_msp.c
```

Verify that the required Timer peripheral and GPIO pin are initialized correctly.

Check:

- Timer clock is enabled.
- GPIO clock is enabled.
- PWM GPIO pin is initialized.
- GPIO is configured as **Alternate Function**.
- Correct Alternate Function is selected.
- Correct Timer Channel is selected.
- Physical PWM output pin matches the configured Timer Channel.
- `HAL_TIM_PWM_Start()` is called.
- Timer Prescaler and Counter Period are correctly configured.

The PWM GPIO should typically be configured as:

```
GPIO Mode → Alternate Function
Alternate Function → Correct TIMx_CHy
```

For example:

```
TIM2_CH1 → Correct GPIO pin with TIM2 Alternate Function
```

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6. PWM Signal Summary

For a 50 Hz ESC PWM signal:

```
PWM Frequency = 50 Hz
PWM Period    = 20 ms
Timer Tick    = 1 µs
Counter Period = 19999
```

Typical ESC pulse values:

```
1000 µs → Minimum throttle
1500 µs → Mid throttle
2000 µs → Maximum throttle
```

The PWM pulse width is controlled by changing the Timer Compare value:

```
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, pulse_width);
```

The overall operation is:

```
STM32
   |
   | Timer
   ▼
PWM Generation
   |
   | 50 Hz
   |
   ├── 1000 µs → Minimum throttle
   ├── 1500 µs → Mid throttle
   └── 2000 µs → Maximum throttle
   |
   ▼
ESC
   |
   ▼
Motor
```

> **Safety:** Always remove the propeller before testing a motor/ESC. Ensure the ESC is properly calibrated and the PWM signal is within the range supported by the ESC.